

E-Commerce Sales Analytics

Project Objective

The primary objective of this project is to develop an interactive E-Commerce Sales & Customer Analytics Dashboard using Power BI to analyze sales performance, product performance, customer behavior, and order trends.

Business Problems

The e-commerce business generates a large amount of sales, product, and customer data. However, without proper analysis and visualization, it can be difficult for management to identify trends, understand customer behavior, and evaluate business performance.

- **Lack of Clear Sales Performance Visibility**
- **Difficulty Identifying Sales Trends**
- **Identifying High-Performing Products**
- **Limited Understanding of Customer Behavior**
- **Difficulty Segmenting Customers**
- **Lack of Geographic Sales Insights**
- **Order and Cancellation Monitoring**
- **Payment Method Analysis**
- **Delivery Performance Monitoring**
- **Difficulty Making Data-Driven Decisions**

Target Audience

The primary users of the proposed e-commerce analytics solution are:

- **Business Managers** - Monitor overall performance, sales, revenue and business growth.
- **Sales Managers** - Analyze sales trend, orders, products, categories and regions
- **Marketing Managers** - Evaluate customer behavior, customer segments and purchasing patterns.
- **Product Managers** - Identify top/low-performing products, categories and brands.

- **Operation Managers** - Monitor order status, delivery, shipping and operational performance.
- **Data Analysts** - Perform detailed analysis and generate actionable business insights from the dashboard.

Scope

The scope of the project is to develop an interactive power BI-based e-commerce analytical solution such as

1. Sales Performance Analysis
2. Customer Analysis
3. Product & Category analysis
4. Geographical Analysis
5. Order & Delivery Analysis
6. Business Insights & Decision Support

KPIs

1. Total Sales/Revenue
2. Total Orders
3. Total Quantity Sold
4. Average Order Value
5. Sales Growth %
6. Total Customers
7. Total Discount
8. Cancellation Rate
9. Return Rate
10. Average Delivery Days

Timeline

Day	Task
Day 1	Dataset understanding
Day 2	Dataset assessment, Data quality checks and cleaning
Day 3	KPI definition and Dashboard mockups
Day 4	Data modeling and DAX foundations
Day 5	Advanced DAX and Dashboard buildings
Day 6	Dashboard testing and Finalization
Day 7	Analysis report and Final submission

Assumptions:

It is assumed that the available e-commerce datasets are sufficiently accurate, complete, and representative of the business activities. Customer, product, order, and transaction information are assumed to be suitable for establishing relationships between the datasets. Data-quality issues such as missing values, duplicates, and inconsistencies are assumed to be identified and appropriately handled during the data-cleaning process. It is also assumed that the available data is sufficient to calculate the required KPIs and support the analysis of sales, customers, products, orders, and business performance. Finally, it is assumed that users have access to the required Power BI environment for viewing and interacting with the dashboard.